INCH-POUND

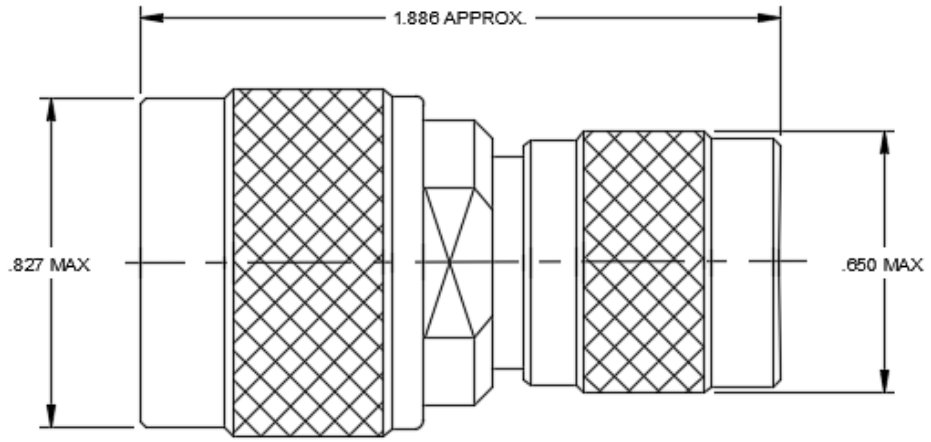
MIL-PRF-55339/57
16 April 2018

PERFORMANCE SPECIFICATION

ADAPTER, CONNECTOR, COAXIAL, RADIO FREQUENCY.
(BETWEEN SERIES N PLUG TO SERIES TNC PLUG), CLASS 2, STRAIGHT PLUG

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall
consist of this specification sheet and MIL-PRF-55339.



Reference	Series	Contact
A	N	Pin
B	TNC	Pin

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. All undimensioned pictorial configurations are for reference purposes only.
3. Interfaces shall be in accordance with MIL-STD-348.

FIGURE 1. General configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:08Z
Check the source to verify that this is the current version before use.

DESIGN AND CONSTRUCTION:

General configuration: See figure 1.

Impedance: 50 ohms, nominal.

Working voltage:

Sea level: 500 V rms

70,000 feet (4.437 kPa): 125 Vrms

Frequency range: 5 to 11 GHz.

Temperature range: -65°C to +165°C.

PERFORMANCE: (Installation torque of 6 to 10 in. lb (0.68Nm to 1.13 Nm), series N).

Dimensions and configuration: See MIL-STD-348 and figure 1.

Center contact retention:	Series TNC	Series N
Axial force (lb, minimum)	6 (26.69 N)	6 (26.69 N)
Torque (in. oz, minimum)	N/A	N/A

Force to engage and disengage:	Series TNC	Series N
Longitudinal force (lb, maximum)	3.2 (14.23 N)	N/A
Torque (in, lb, maximum)	2 (0.22 Nm)	6.0 (0.68 Nm)

Coupling proof torque: 15 in. lb (1.69 Nm), minimum, series N.

Mating characteristics:

Outer contact, series TNC:

Min test ring ID - .319 in. (8.10 mm), max.

Pin finish – 16 microinches (0.406 μ m)

Insertion force – 5 lb (22.24 N), max.

Insertion depth - .093 in. (2.36 mm), min.

No. of insertions – 1.

Max test ring ID - .324 in. (8.23 mm), min.

Test ring finish – Not applicable.

Insertion depth - .031 in. (0.79 mm), max.

Number of insertions – 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:08Z
Check the source to verify that this is the current version before use.

Outer contact, series N:

Minimum test ring ID: .316 inch (8.03 mm), maximum.

PIN finish: 16 microinches (.406 µm).

Insertion force: 25 lb (111.20 N), maximum.

Insertion depth: .093 inch (2.36 mm), minimum.

Number of insertions: 1.

Slotted member contacts only, series N.

Maximum test ring ID: .324 inch (8.23 mm), minimum.

Test ring finish: 16 microinches (.406 µm).

Insertion depth: .031 inch (0.79 mm), maximum.

Number of insertions: 1.

Permeability: <2.0

Seal:

Hermetic: Not applicable.

Pressurized: Not applicable.

Weatherproof: Not applicable.

Insulation resistance: 5,000 megohms, minimum.

VSWR: 1.35:1, maximum at .5 to 9 GHz. 1.5 maximum at 9 to 11 GHz.

RF leakage (total): -60dB, minimum, 2 to 3 GHz.

RF insertion loss: $.060\sqrt{F(GHz)}$ dB maximum, tested at 6 GHz.

Durability: 500 cycles minimum at 12 cycles/minimum maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Dielectric withstanding:

Test voltage: 1,500 Vrms, minimum (sea level).

Contact resistance (milliohms, maximum).

Contact	Initial	After
Center	2.0	2.5
Outer	.25	N/A
Outer (-70001)	.5	N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:08Z
Check the source to verify that this is the current version before use.

Vibration, high frequency:

Interruptions: 1 µs, maximum.

Shock: Test condition I.

Thermal shock: Test condition C.

Moisture resistance: 200 megohms, minimum.

Corona level:

Voltage: 375 V, minimum.

Altitude: 70,000 feet (4.437 kPa), minimum.

RF high potential withstanding voltage:

RF voltage: 1,000 Vrms, minimum.

Frequency: 5 MHz, minimum.

Salt spray (corrosion): Applicable.

Coupling mechanism retention force: 100 lb (444.82 N), minimum.

Part or Identifying Number (PIN). M55339/57-00001

Referenced documents: In addition to MIL-PRF-55339, this document references the following:

MIL-STD-348

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2018-038)

Review activities:
Army - AR, AT, EA, MI
Navy - AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:08Z
Check the source to verify that this is the current version before use.